

The other is more important

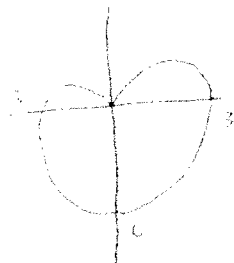
Calculus II

Worksheet

Name:

1. Graph the function $r = 3(1 - \sin(\theta))$.

r	θ
3	0
0	$\pi/2$
3	π
6	$3\pi/2$

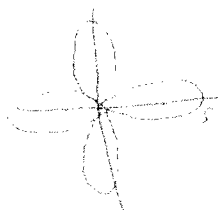


2. Set up an integral to find the area enclosed by $r = 3(1 - \sin(\theta))$.

$$\int_0^{2\pi} \frac{1}{2} (3(1 - \sin(\theta)))^2 d\theta$$

3. Find the area enclosed by three leaves of the four-leafed rose $r = 5 \cos(2\theta)$.

r	θ
5	0
0	$\pi/4$
-5	$\pi/2$
0	$3\pi/4$



$$3 \int_{-\pi/4}^{\pi/4} \frac{1}{2} (5 \cos(2\theta))^2 d\theta \quad \text{[by symmetry]} \quad 3 \int_0^{\pi/4} \frac{1}{2} 25 \cos^2(2\theta) d\theta$$

4. Evaluate $\int \cos^2(3x) dx$.

Recall $\cos^2 \theta = \frac{1}{2} (1 + \cos(2\theta))$. So

$$\int \cos^2(3x) dx = \int \frac{1}{2} (1 + \cos(6x)) dx = \frac{1}{2} \left(x + \frac{1}{6} \sin(6x) \right) + C$$

5. Set up an integral to find the length of the curve $y = 4 \ln x$, from $x = 1$ to $x = 3$

(a) By integrating with respect to x .

$$\int_1^3 \sqrt{1 + \left(\frac{4}{x}\right)^2} dx$$

(b) By integrating with respect to y .

$$y = 4 \ln x \quad \text{so} \quad x = e^{\frac{1}{4}y}$$

$$\int_0^{4 \ln 3} \sqrt{1 + \left(\frac{1}{4} e^{\frac{1}{4}y}\right)^2} dy$$

6. Set up an integral to find the area of the surface generated by rotating the curve $y = 4 \ln x$, from $x = 1$ to $x = 3$

(a) About the x -axis.

$$\int_1^3 2\pi (4 \ln x) \sqrt{1 + \left(\frac{4}{x}\right)^2} dx$$

(b) About the line $y = -3$.

$$\int_1^3 2\pi (4 \ln x - (-3)) \sqrt{1 + \left(\frac{4}{x}\right)^2} dx$$

(c) About the line $x = -3$.

$$\int_0^{4 \ln 3} 2\pi (e^{\frac{1}{4}y} - (-3)) \sqrt{1 + \left(\frac{1}{4} e^{\frac{1}{4}y}\right)^2} dy$$

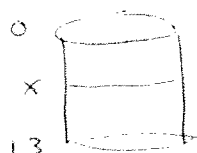
Coordinate systems may vary
 ↓
 Calculus II

See my power point for details on how I work these.

Worksheet

Name:

Copyright 2006 Dr. Brock Wilbanks and Texas Tech University. Unauthorized reproduction prohibited.



1. A cylindrical tank has a radius of 5 ft and a height of 13 ft. Set up an integral to determine how much work is done in emptying it if

(a) It is filled with molasses (which weighs 235 lb/ft³).

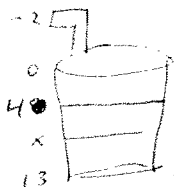
$$\int_0^{13} \underbrace{x}_{\text{distance}} \underbrace{235 \pi (5)^2 dx}_{\text{Force}}$$

(b) It is filled to a depth of 9 ft and the molasses must be pumped out a 2 ft pipe at the top of the tank.

$$\int_4^{13} (x+2) 235 \pi (5)^2 dx$$

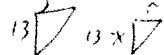
[0']

$$\int_6^{15} x 235 \pi (5)^2 dx \quad \text{etc.}$$



2. A conical tank has a radius of 5 ft and a height of 13 ft. Set up an integral to determine how much work is done in emptying it if

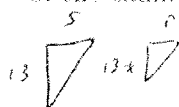
(a) It is filled with milk (which weighs 64.5 lb/ft³).



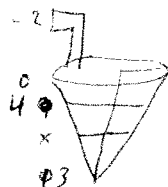
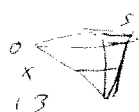
$$\frac{5}{13} = \frac{r}{13-x}$$

$$\int_0^{13} \underbrace{x}_{\text{distance}} \underbrace{64.5 \pi \left(\frac{5}{13} (13-x) \right)^2 dx}_{\text{Force}}$$

(b) It is filled to a depth of 9 ft and the milk must be pumped out a 2 ft pipe at the top of the tank.



$$\int_4^{13} (x+2) 64.5 \pi \left(\frac{5}{13} (13-x) \right)^2 dx$$



3. Discuss the following completely.

(a) What is the fundamental idea of Chapter 6? See Big Idea slide

They should be able to completely explain with full sentences

(b) Suppose an object moves along the x -axis from $x = 2$ to $x = 5$ and is acted on by a force $F(x) = x^2$. Set up an integral to determine the amount of work done on the particle.

$$\int_2^5 x^2 dx$$

(c) Use the fundamental idea of Chapter 6 to explain in complete detail why the work on the particle is given by your integral.

Very Important Complete sentences. Full details

Apply part (b) to the changing force to get work.

Big Idea

Integration

1. Something was changing, so we couldn't use the old algebra formulas.
2. Break the problem into pieces.
3. Pretend everything is constant on each piece.
4. Add up the pieces. (This is called a *Riemann Sum*)

$$\sum f(t) \Delta t$$

5. If we use more and more pieces, the limit is the right answer! (This limit is a *definite integral*.)

$$\int_0^3 f(t) dt$$